

PhonePe Transaction Analysis Dashboard

End-to-End Business Intelligence Project

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Power BI Data Analytics Project

Power BI • Excel • Power Query • DAX

Project Overview

This project focuses on analyzing digital payment transaction data for a PhonePe-like platform using Power BI. The dashboard was designed to monitor transaction performance, payment success rates, failed payment reasons, and service-wise business metrics.

The project simulates a real-world Business Intelligence workflow where stakeholders can dynamically explore transaction insights and monitor operational performance using interactive visualizations.

Business Problem Statement

Digital payment platforms process millions of daily transactions across services such as insurance, loans, money transfers, and recharge payments.

Without proper data analysis, it becomes difficult to:

- Monitor transaction performance across services
- Identify high-revenue business segments
- Detect operational issues and failed payments
- Track monthly transaction trends
- Improve customer experience and payment reliability

This project aims to answer key business questions such as:

1. What is the total transaction volume across services?
2. Which services contribute the most revenue?
3. What are the common reasons for failed transactions?
4. How do transaction trends vary over time?
5. How can data insights improve operational efficiency?

Dataset Information

Key Dataset Columns

- User_ID
- User_Name
- Join_Date
- Age

- Transaction_ID
- Service_Type
- Transaction_Amount
- Payment_Status
- Failure_Reason
- Service_Category

Tools and Technologies Used

Tool	Purpose
Power BI Desktop	Dashboard Development and Visualization
Microsoft Excel	Data Cleaning and Validation
Power Query Editor	Data Transformation
DAX	KPI and Calculated Measures

Table 1: Technology Stack

Dashboard Overview

The dashboard contains multiple interactive pages for different PhonePe services.

Features included:

- Date Range Filtering
- KPI Cards
- Monthly Transaction Trends
- Payment Status Analysis
- Failed Payment Insights
- Service-wise Revenue Analysis

Homepage Dashboard

The homepage provides an overview of total transaction amount, successful transactions, failed payments, and service-wise revenue distribution.

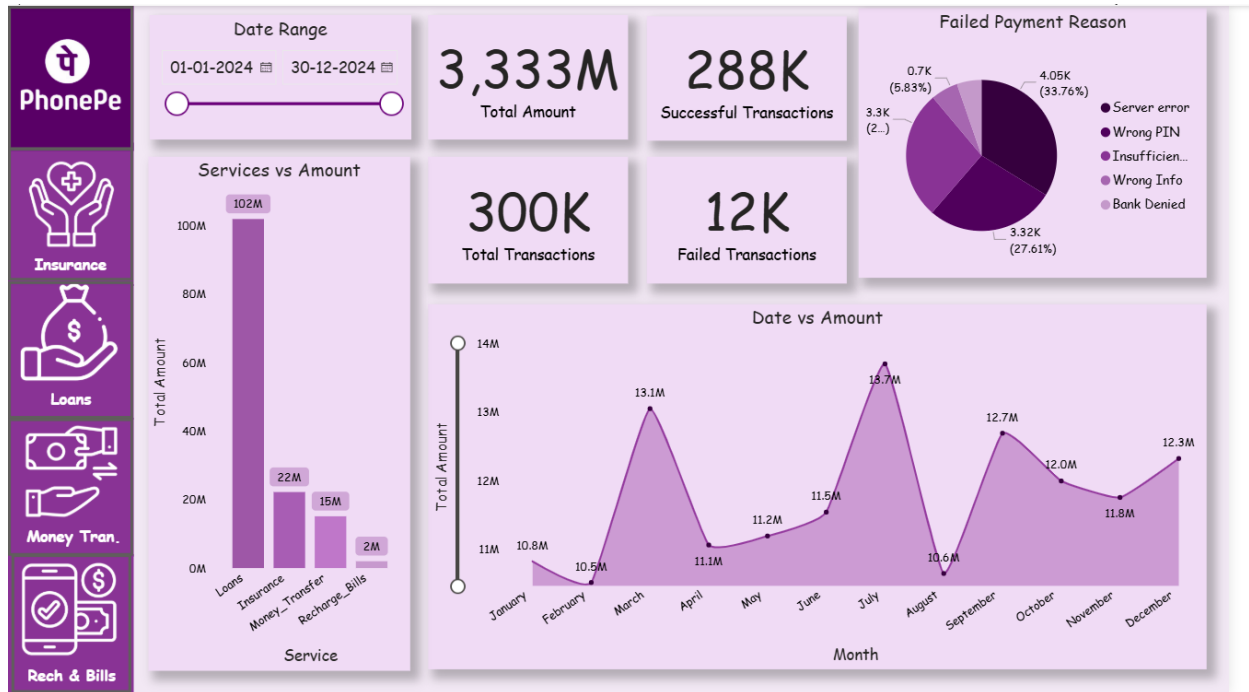


Figure 1: Homepage Dashboard Overview

Insurance Dashboard

The Insurance dashboard visualizes premium collection trends, payment status, and service-wise transaction insights across insurance categories.

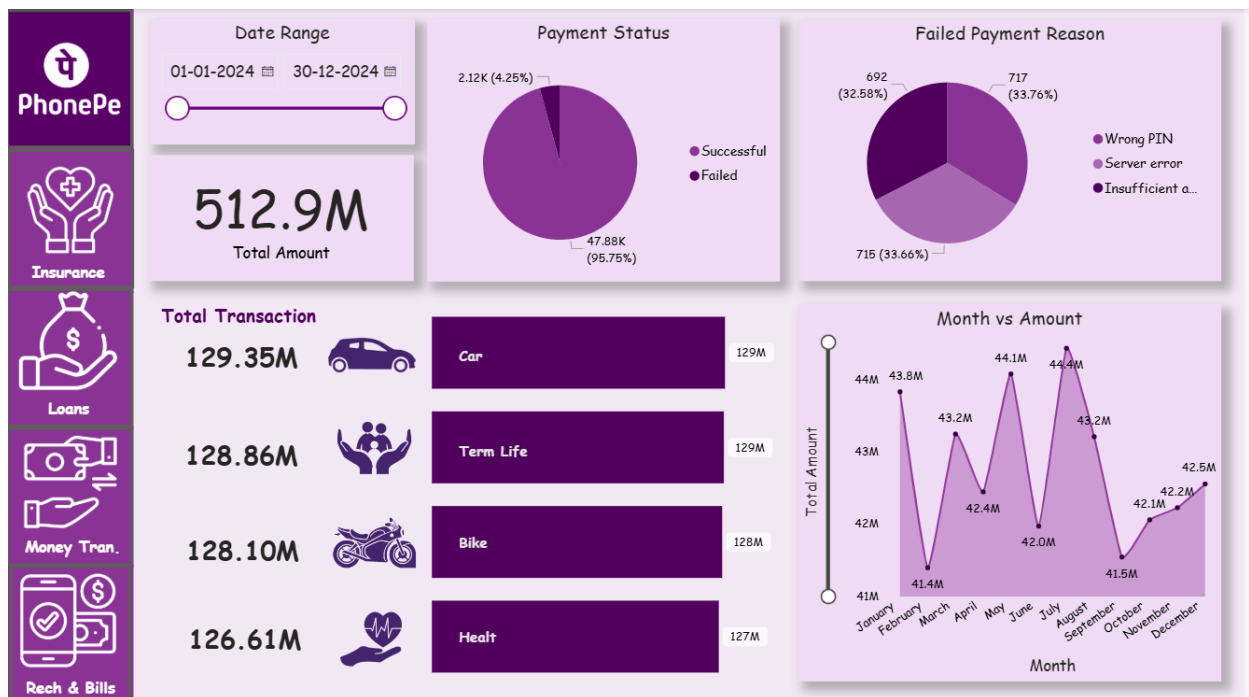


Figure 2: Insurance Service Dashboard

Loan Dashboard

The Loan dashboard tracks loan transaction metrics, failed payment analysis, and monthly transaction trends across multiple loan categories.

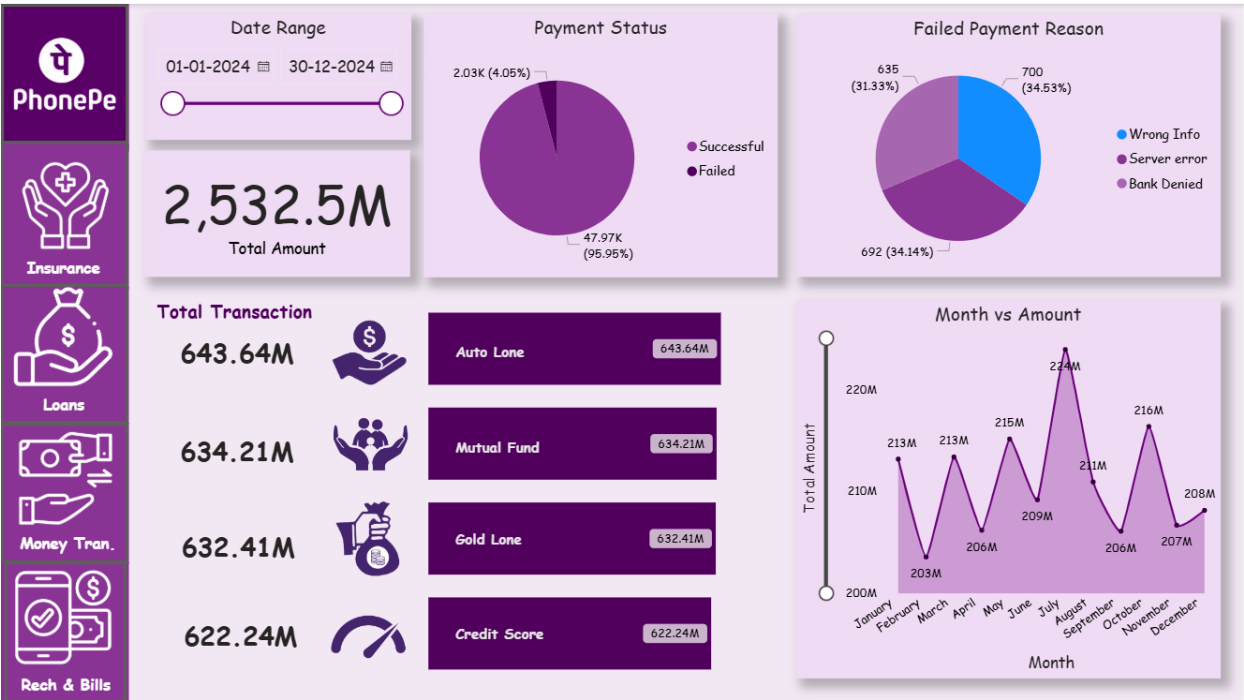


Figure 3: Loan Service Dashboard

Money Transfer Dashboard

The Money Transfer dashboard analyzes transaction volume, payment success rate, and monthly transaction trends for transfer services.

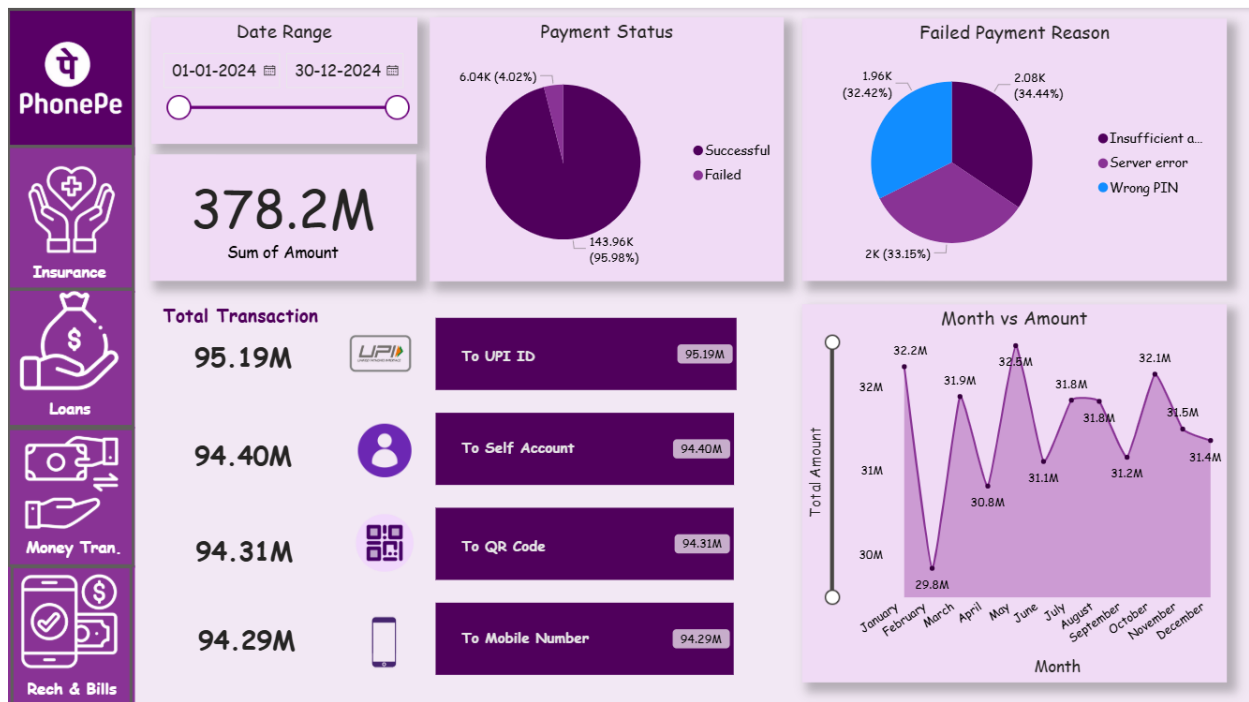


Figure 4: Money Transfer Dashboard

Recharge and Bills Dashboard

The Recharge and Bills dashboard tracks recharge activity, payment categories, transaction amount trends, and failed payment analysis.

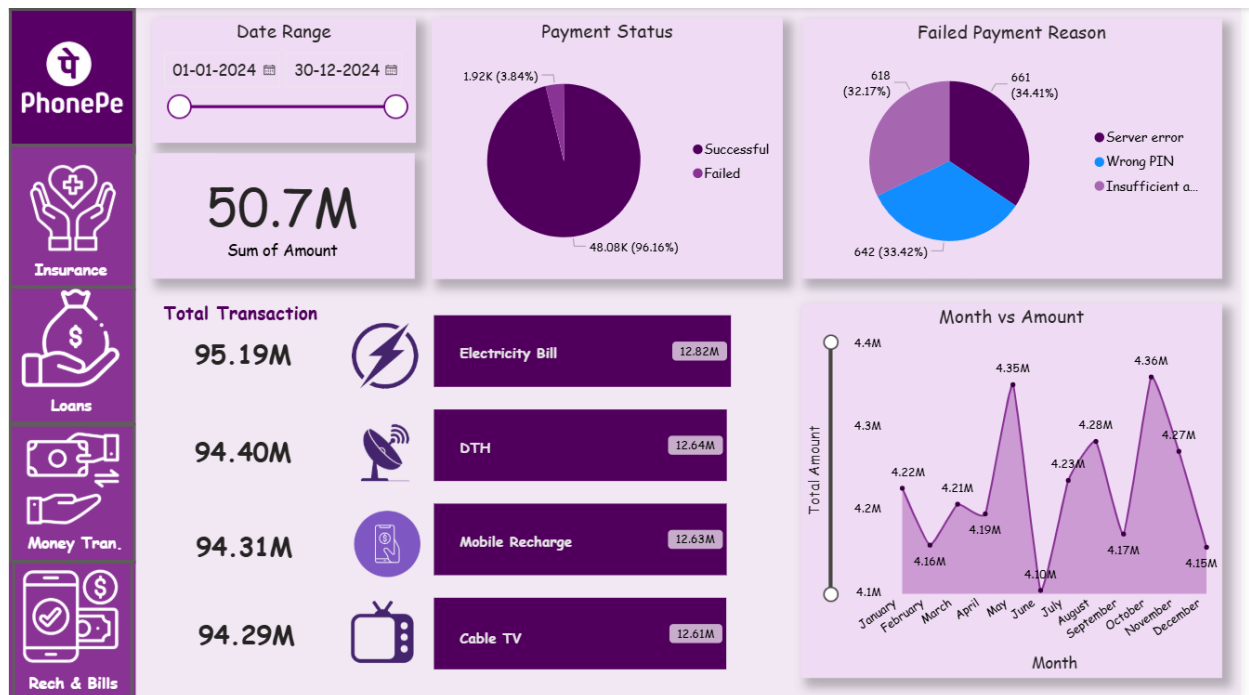


Figure 5: Recharge and Bills Dashboard

Failed Payment Analysis

The dashboard identifies major failed payment reasons including:

- Server Errors
- Wrong Payment Details
- Insufficient Balance
- Wrong PIN

These insights help improve transaction success rates and operational reliability.

Key Insights

- Most transactions were generated through Money Transfer and Recharge services.
- Failed transactions were primarily caused by server issues and insufficient balances.
- Interactive filters enabled dynamic service-level transaction analysis.
- Monthly transaction trends highlighted peak transaction periods across services.
- The PhonePe-inspired color palette improved dashboard consistency and visual storytelling.

Why This Project Matters

This project demonstrates how Business Intelligence tools can transform raw transactional data into actionable insights.

The project showcases:

- Real-world dashboard development
- KPI-driven business reporting
- Interactive data storytelling
- Service-wise operational analysis
- End-to-end Power BI workflow

Conclusion

The PhonePe Transaction Analysis Dashboard demonstrates how Power BI can be used to analyze large-scale digital payment data and uncover meaningful business insights.

From monitoring millions of transactions to identifying failed payment patterns, the project highlights the importance of data visualization and analytics in business decision-making.